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1. **[1-mark]** 1-marks for good handwriting and clear presentation.

2. **[5-mark]** Find Green's function(Laplace equation) on $(0,\infty)\times(0,\infty)$.

2] This is the Quarter Plane. We will be using the method of images.
The Green's function $G(x,y)$ we get actually depends on the type of Boundary for +ve x-axis and +ve y-axis : is it Dirichlet or Neumann.

To factor this in, define:

$$\alpha_{x_1} := \begin{cases} 1 & \text{+ve x axis has Neumann} \\ -1 & \text{+ve x axis has Dirichlet} \end{cases}$$

$$\text{lly, } \alpha_{x_2} := \begin{cases} 1 & \text{+ve Y axis has Neumann} \\ -1 & \text{+ve Y axis has Dirichlet} \end{cases}$$

In the method of images, we put artificial sources at specific points to get the boundary conditions desired. For Quarter Plane, given $y = (x_1, x_2)$, define

$$y^{(1)} := (-x_1, x_2), \quad y^{(2)} := (x_1, -x_2), \quad y^{(3)} := (-x_1, -x_2).$$

Now, for $x, y \in \mathbb{R}^2$,

$$G(x,y) := \Phi(y-x) + \alpha_{x_1}\Phi(y^{(1)}-x) + \alpha_{x_2}\Phi(y^{(2)}-x) + \alpha_{x_1}\alpha_{x_2}\Phi(y^{(3)}-x)$$

where $\Phi(z) := -\frac{1}{2\pi}\ln|z|$ is the fundamental soln in 2D
(the Green's function for unbounded domain)

substitutes definition of Φ and solving:

$$G(x,y) = -\frac{1}{2\pi}\left[\ln\left((|y-x|)(y^{(1)}-x)^{\alpha_{x_1}}(y^{(2)}-x)^{\alpha_{x_2}}(y^{(3)}-x)^{\alpha_{x_1}\alpha_{x_2}}\right)\right]$$

3 **6-mark** Write down a solution of: $u_t = u_{xx}$ on $(x,t)\in(0,1)\times(0,\infty)$, $u(x,0)=\sin(x)$ for $x\in[0,1]$.

Since we can write down any solution, I will find a solution satisfy 2 additional constraints:
$u(0,t)=0$ and $u_x(1,t)=0$. $\forall t \in [0,\infty)$ (calculated on $[0,1]\times[0,\infty)$ then restrict the domain later.

Let's use Separation of Variables: Assume $u(x,t) = X(x)T(t)$ where $X:[0,1]\to\mathbb{R}$ and $T:[0,\infty)\to\mathbb{R}$.

$u_x = X'T$, $u_{xx} = X''T + 0$, $u_t = T'X$.

$\therefore u_t = u_{xx} \Rightarrow \frac{T'}{T} = \frac{X''}{X} \overset{\text{let}}{=} -\lambda$ for some $\lambda\in\mathbb{R}$.

We solve $X'' = -\lambda X$:

Case 1: $\lambda<0$, let $\lambda = -\omega^2$:

$\Rightarrow X'' = \omega^2 X$. Characteristic eqn $\Rightarrow r^2 = \omega^2 \Rightarrow r = \pm\omega$ (discrete real roots)

$\therefore X(x) = C_1 e^{\omega x} + C_2 e^{-\omega x}$

putting in constraints:

$u(0,t) = 0 \Rightarrow X(0)T(t) = 0 \Rightarrow X(0) = 0$ ($\because T(t)=0 \Rightarrow u=0 \Rightarrow\Leftarrow$ as $u(x,0)=\sin x$)

$X(0) = C_1e^0 + C_2e^0 = C_1 + C_2 = 0 \Rightarrow C_1 = -C_2$ $\therefore X(x) = C_1(e^{\omega x} - e^{-\omega x})$

$u_x(1,t) = 0 \Rightarrow X'(1)T(t) = 0$ $\because T \neq 0$, $X'(1) = 0$.

$\Rightarrow C_1\omega(e^{\omega} + e^{-\omega}) = 0 \Rightarrow C_1 = 0$. $\therefore$ trivial solution.
(or $\lambda = 0$ not in case)

Case 2: $\lambda = 0$,

$X'' = 0 \Rightarrow X = C_1x + C_2$, $X'(x) = C_1$

$X(0) = 0 = C_2$
$X'(1) = C_1 = 0$ $\therefore$ trivial solution only.

Case 3: $\lambda>0$, $\lambda = \omega^2$

$X'' = -\omega^2 X$. Char eqn: $r^2 = -\omega^2 \Rightarrow r = \pm i\omega$.

$\therefore X(x) = C_1\cos(\omega x) + C_2\sin(\omega x)$

$X(0) = 0 = C_1$
$X'(x) = -C_1\omega\sin(\omega x) + C_2\omega\cos(\omega x)$
$X'(0) = C_2\omega = 0 \Rightarrow C_2 = 0$ trivial.

$\cancel{\nabla u \cdot \nabla u = \operatorname{div}(u\nabla u) - u\Delta u}$

$\int \nabla u \cdot \nabla u$

$\int_\Omega \operatorname{div}(u\nabla u) = +\int_{\partial\Omega} u \nabla u - \int_\Omega u\Delta u.$

$\int_\Omega |\nabla u|^2 \cancel{+\int_\Omega u\Delta u}$

4 **[6-mark]** Prove that for all $u \in C_c^\infty((0,1)\times(0,1))$, $\int_{(0,1)\times(0,1)} |\nabla u|^2 \geq \int_{(0,1)\times(0,1)} u^2$.

$+ u_x = 0$

$X'(1) = 0$

4 **[6-mark]** Let $f \in C_c^2(\mathbb{R}^3)$. Then prove that any bounded solution of $-\Delta u = f$ in $\mathbb{R}^3$ has the form $u(x) = \int_{\mathbb{R}^3} \Phi(x-y)f(y)dy$ $+C$ for some constant $C \in \mathbb{R}$. Φ is the fundamental solution of Laplace equation.

This comes from Green's integral formula:

$$u(x) = \int_\Omega G(x,y)\, f(y)\, dy - \int_{\partial\Omega} \frac{\partial G(x,y)}{\partial\nu} g(s)\, ds$$

where $u(s) = g(s)$ on $\underset{s\in}{\partial\Omega}$, where $f = -\Delta u$.

~~Laplace equation, f = 0 ⟹ g = 0.~~

Also, $G(x,y) = \Phi(y-x) + \varphi^x(y)$ where $\varphi^x(y)$ is corrective term to enforce Boundary condition.

Take $\Omega \subseteq \mathbb{R}^3$. & $\partial\Omega$ exists.

$$\therefore\ u(x) = \int_\Omega [\Phi(y-x) + \varphi^x(y)]\, f(y)\, dy - \underbrace{\int_{\partial\Omega} \frac{\partial G(x,y)}{\partial\nu} g(s)\, ds}_{C_1 \in \mathbb{R}}$$

$$= \int_\Omega \Phi(y-x) f(y)\, dy + \underbrace{\int_\Omega \varphi^x(y) f(y)\, dy}_{C_2 \in \mathbb{R}} - C_1$$

$$= \int_{\mathbb{R}^3} \Phi(y-x) f(y)\, dy - \underbrace{\int_{\mathbb{R}^3\setminus\Omega} \Phi(y-x) f(y)\, dy}_{C_3 \in \mathbb{R}} - C_2 - C_1$$

$$= \int_{\mathbb{R}^3} \Phi(y-x) f(y)\, dy - C_3 - C_2 - C_1$$

Hence proved.

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